STRATEGIC MATERIALS POLICY TRACKER

RESEARCH DOSSIER 001

Tungsten Supply Security

Policy Mechanisms and Evidence of Implementation

Selected Measures in China, the EU, the UK and Japan

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Independent comparative research

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About this research

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This study asks what the documentary record establishes about the implementation of selected tungsten policy measures in China, the European Union, the United Kingdom and Japan. The unit of comparison is the instrument and its targeted supply-chain function. The comparison evaluates documented implementation of selected instruments through their legal, financial, industrial and commercial effects.

Implementation differs materially across the selected cases. Hemerdon supplies financial and operating disclosures, the Japanese cases document grant awards for planned conversion capacity, and EU recognition covers projects at different development stages. Hemerdon provides evidence of a drawn public loan and reported processing. Japanese projects reveal awarded support for future conversion capacity. European recognition covers projects with sharply different development positions.

The analysis combines the public SMPT dataset retrieved on 3 November with supplemental official documents, corporate disclosures and financial statements. The dataset matches repository commit 4fc59c7ea38ca788eb7537353274a953c90e13eb, which merged PR #188. The dataset revision date identifies the published corpus version. Project-specific verification dates appear in the claim-level evidence record.

Verification relies on the cited public records, official legal materials and issuer disclosures; physical inspections and independent bank confirmation remain additional verification steps. The research cutoff remains 3 November 2026.

Reading route. The executive summary sets out the answer. The evidence chapters move from legal intervention to public capital and implementation. The final sections explain commercial implications, the SMPT research workflow, reproducibility and source limitations. Bracketed source identifiers refer to the reference register at the end.

THE ANSWER

Executive summary

The selected documents establish implementation at several distinct levels: legal obligations, project recognition, public-loan drawdown, reported processing and a commercial supply agreement. Their effects require separate legal, financial, industrial and delivery metrics. At Hemerdon, the accounts disclose a fully drawn GBP 25 million NWF loan by approval on 2 September 2026. Processing is company-reported, and September disclosures add an Elmet offtake agreement. Sustained delivery records and incremental output attributable to public support are the next verification requirements. [S21, S31, S35, S36]

First, the UK announced a financing package valued at up to GBP 71 million. The package comprises GBP 36 million of equity, a GBP 25 million core loan and a GBP 10 million accordion facility subject to subsequent commitment. The government’s opportunity to negotiate purchases of up to half the specified tungsten production is a separate, unpriced arrangement. Counting the package and its components would double-count the same funding; treating the accordion as paid capital would overstate execution. [S14, S20, S31]

Second, public recognition is compatible with very different industrial positions. El Moto’s developer reports construction underway. Its separately recognized EU APT Tungsten Plant remains at the evaluation stage, with an investment decision still pending. Strategic Project status confers administrative recognition; operational and financing milestones require project-level verification. [S04, S05, S08, S09, S24, S25]

Third, Japan’s selected projects target conversion capability. Allied Material plans to expand powder and carbide capacity; Japan New Metals plans more scrap-to-APT processing at Akita. JOGMEC now lists grant decisions of approximately JPY 7.5 billion and JPY 2 billion respectively. The recorded milestone is the grant decision. Disbursement and output from the supported expansions are later milestones. Their operating targets fall in 2028–29. [S12, S13, S17]

Fourth, Chinese export licensing should be assessed against its own mechanism. Announcement No. 10 regulates specified tungsten items and production technology, with technical conditions for some categories. It establishes a legal transaction boundary. Evaluation of transactional effects requires item-aligned licensing, enforcement and shipment series beyond the present dataset. [S01, S19]

The commercially useful finding is that supply security requires a connection between the product produced, conversion access and a customer’s requirements. Hemerdon’s disclosed offtake adds a route to allied processing, while Japanese support targets conversion capacity itself. A qualified supply chain additionally requires suitable conversion, customer acceptance and dependable deliveries. Monitoring should follow the specific missing link for each instrument.

COMPARATIVE FRAMEWORK

Four policy instruments address distinct supply-chain constraints

A government can change the legal conditions under which material moves, improve a project's treatment within administrative processes, or finance assets that may eventually produce material. These actions affect supply security through different channels. The comparison assesses each policy against its actual mechanism and corresponding evidence.

Legal implementation of a licensing rule occurs through regulated transactions. A designation can matter before any public money moves. A loan can be drawn before the financed expansion operates. Each observation answers a different question. The analysis connects established dimensions and identifies the next records required for the remaining links.

The jurisdictions also overlap. Hemerdon is located in the UK and receives UK financing, but it was also included in the EU's June 2025 third-country Strategic Projects decision. These records describe one physical project receiving attention under two policy frameworks. Policy origin, project location and capital provider therefore remain separate fields. [S06, S07, S14]

Exhibit 1. Four instruments require different evidence of implementation

System	Instrument	What is established	What remains to test
China	Export licensing	Permission for defined goods and technology	Approvals, enforcement and item-aligned shipments
EU	Strategic Projects	Administrative recognition; development support	Financing closure, permit progress, construction and output
UK	Public equity / loan	Capital for Hemerdon restart; loan drawn	Expenditure, sustained output, delivery and additionality
Japan	Certification / grants	Awards supporting conversion expansion	Payments, equipment installation, commissioning and new output

Sources: S01, S04–S09, S12–S14, S20, S21, S27, S31. The study covers selected mechanisms with distinct implementation measures.

ECONOMIC CONTEXT AND MECHANISMS

China supplies approximately 79% of mined tungsten

USGS estimates 2025 world mine production at 85,000 tonnes of tungsten content, including 67,000 tonnes in China: approximately 79% using the rounded estimates. The 79% estimate concerns mined tungsten content. Downstream product shares and buyer-level exposure require separate data. USGS also identifies limits to substitution in several applications, including potential performance or cost penalties. These features make tungsten a useful case for examining stage-specific dependencies. [S32, tungsten chapter pp. 200–201]

The four systems were chosen to examine contrasting instruments already represented in the frozen corpus. The sample is purposive: four jurisdictions with contrasting instruments already represented in the frozen corpus. The United States, Canada and other producers are outside this comparison. China is examined as the issuer of a licensing measure; the other cases concern development or supply assurance. Each measure is assessed against its stated or observable policy mechanism.

Licensing can increase transaction uncertainty or delay market access, influencing buyers' inventory and supplier choices. EU recognition can reduce administrative coordination costs and permit uncertainty, potentially improving investment timing. These mechanisms suggest tests of licensing delay and project-level permitting acceleration, using administrative transaction and permit data.

UK equity and lending provide capital but allocate different risks and returns. Equity can absorb losses; debt creates repayment and refinancing obligations. Japanese grants can reduce eligible investment costs and influence scale or timing. The incremental effect on private financing and investment timing requires a counterfactual financing assessment. Gross financing and fiscal subsidy cost are distinct financial concepts. A welfare or opportunity-cost comparison would require additional fiscal and market data.

Resilience is defined here as the ability to maintain usable supply through disruption. More geographically diverse capacity may contribute, but feedstock, conversion, quality qualification, delivery commitments and exposure to common shocks still matter. The cases identify specific supply-chain dependencies for subsequent resilience measurement.

INDUSTRIAL FUNCTION

Tungsten security depends on conversion and usable products

A mine, an APT plant and a powder producer supply different products. Mining and concentration create an upstream feedstock. Chemical conversion can produce ammonium paratungstate, an intermediate commonly abbreviated APT. Reduction and carburisation produce tungsten and tungsten carbide powders used further along industrial chains. Recycling can return material to several conversion steps. The schematic depicts the production and conversion functions described in the selected sources. Actual shipment routes require separate transaction data. [S01, S13, S17, S25]

The selected UK financing supports a mine restart and processing infrastructure. The Japanese initiatives address conversion and recycled feedstock. The October EU list adds recognition for upstream extraction, an APT proposal and a recycling project. China’s controlled item list reaches intermediate products, specified solid forms and alloys, and associated technology. These functions create distinct physical capabilities and commercial dependencies. [S01, S08, S14]

This changes the interpretation of diversification. Additional geographically diverse concentrate still requires suitable chemical conversion. Additional powder-making equipment increases processing options when compatible feedstock is available. A recycling expansion may replace some imported intermediates while retaining dependencies on other inputs. Those are distinct capabilities with different commercial dependencies.

Exhibit 2. Documented interventions address different supply-chain functions

Industrial function	Selected measures and projects
Extraction / concentration	Hemerdon restart financing; EU recognition for El Moto, P6, Borralha and Hemerdon
Chemical conversion / APT	Japan New Metals scrap-to-APT expansion; proposed EU APT plant; China controls APT and oxides
Powders / carbide / alloys	Allied Material reduction and carburisation expansion; specified Chinese carbide, solid forms, alloys and technology
Recovery / recycling	REW-ITA green HUB; Japan New Metals scrap feedstock. Recycling can enter different conversion steps.

Sources: S01, S04, S06, S08, S13, S14, S17. Functional schematic of selected supply-chain stages. Shipment routes and quantities require separate transaction evidence. As of 3 November 2026.

HEMERDON TIMELINE

Hemerdon activity began before the August financing announcement

This chronology distinguishes announcements, activity and disclosure dates. July is recorded at month precision. The 2 September accounts-approval date establishes the latest date by which the core loan had been fully drawn. The precise transfer date requires a transaction record. The full register preserves all 16 observations.

Some concentrate activity preceded the NWF announcement, but anticipated finance could already have influenced decisions. Different commissioning stages also have different disclosed dates. The chronology establishes the sequence of publicly reported milestones. Attribution requires evidence about the restart trajectory under an alternative financing scenario. [S20, S31]

Exhibit 3. Selected interventions and implementation disclosures, 2025–26

Date	Observation	Action / disclosure
2025-02-04	Control	China: Item-specific tungsten licensing
2025-03-25	Designation	EU: El Moto and P6 recognized
2025-06-04	Designation	EU / UK: Tungsten West recognized
2026-07	Activity month	UK: Concentrate activity reported; month precision
2026-08-25	Finance	UK: NWF financing terms announced
2026-09-02	Execution bound	UK: Loan fully drawn by accounts approval
2026-09-09	Project plan	Japan: Akita recycling expansion disclosed
2026-09-15	Operation report	UK: Ore processing / ramp-up reported
2026-09-23	Commercial contract	UK: Elmet binding offtake announced; deliveries unverified
2026-10-09	Designation	EU: APT, Borralha, REW-ITA decision

Sources: timeline.csv, which retains all 16 research chronology entries and their source IDs. Ten pivotal observations displayed; July has month precision. Disclosure dates and activity dates are separate categories; 2 September bounds the loan drawdown. Selective coverage.

LEGAL INTERVENTION

China requires export licences for specified tungsten products

China’s 4 February 2025 Announcement No. 10 requires exporters of the listed items to seek permission under the export-control framework. For tungsten, the item text includes APT, tungsten oxides and tungsten carbide outside a separately controlled category. Other entries cover solid tungsten forms and alloys subject to stated specifications, plus production technology and technical information. The operative item descriptions and their specifications determine coverage; customs commodity numbers serve as references. [S01, S19]

The distinction matters because “tungsten exports are controlled” is too broad to serve as an item-level finding. For example, the solid-tungsten entry combines composition or material conditions with the ability to machine specified shapes and dimensions. The alloy entry contains several physical-property thresholds. Shipment-level classification requires the full Chinese item descriptions, technical thresholds and transaction details. The analysis uses the original Chinese text to describe the instrument. Actual shipment classification requires a separate legal and technical assessment.

SMPT separates the licensing clause from a May 2025 anti-smuggling campaign. The latter addresses enforcement of existing strategic-mineral controls, including misdeclaration and transshipment practices. It is an enforcement campaign addressing existing strategic-mineral export controls. Counting the two records as equivalent restrictions would obscure the difference between a legal requirement and administrative enforcement. [S02]

The legal obligation is documented. Assessing trade and industrial outcomes requires aligned transaction and production series. A persuasive effectiveness assessment would require defined observations on applications, approvals, denials, processing times, enforcement or actual shipments, with product coverage aligned to the rule. Broad trade headings can aggregate products with different control status; item-level series are necessary for attribution.

The appropriate conclusion is therefore specific: the selected measure creates a licensing obligation for defined goods and technology. Effectiveness assessment therefore requires evidence of licensing decisions, enforcement outcomes and related trade flows. Regulated export transactions are the immediate implementation test for this instrument. Its immediate implementation takes place at a regulated transaction boundary.

ADMINISTRATIVE SUPPORT

EU Strategic Project status provides permitting and coordination support

The EU's March 2025 list recognizes El Moto and P6 Metals. The June third-country list recognizes Tungsten West. The October 2026 decision adds the EU APT Tungsten Plant, Borralha and REW-ITA green HUB. Together these make six selected tungsten designation records under the EU scheme. The selected projects span extraction, processing and recycling, each with a different industrial function. [S04–S09]

CRMA Article 11 sets default permit-process limits for Strategic Projects in the Union of 27 months for extraction and 15 months for processing or recycling. Projects already in permitting and qualifying extensions have 24- and 12-month limits after recognition. The provision excludes specified environmental-assessment preparation time and permits defined extensions. The statutory permitting periods operate under Article 11's specified triggers, exclusions and extensions. They concern permit processing. Hemerdon, located in the UK, falls outside the territorial scope of the EU domestic permitting provisions. [S33, Article 11; S27]

The legal distinction remains visible in the October decision. Article 1 recognizes the annexed projects; the annex identifies the project and relevant raw materials. The annex describes project recognition. Plant operating status must be established from industrial records. The APT developer identifies the project as awaiting its final investment decision. The two statements concern different administrative and commercial milestones. [S08, S09, S25]

The October decision was adopted on 9 October and published in the Official Journal on 22 October. Article 2 addresses it to the listed promoters. TFEU Article 297(2) makes notification relevant to the effect of an addressed decision. The verified dates are 9 October adoption and 22 October Official Journal publication. The notification date remains a separate legal verification requirement. The adopted recognition text is established by the official decision. [S09, S30, S34]

Designation counts can therefore reveal where a government has selected projects for attention. Capital commitments, permit processing times and operating prospects require separate project-level evidence. Those questions require different evidence.

PROJECT READINESS

El Moto advances construction as the APT proposal awaits investment approval

Abenójar’s two projects are particularly informative because the developer reports different industrial positions under the same EU designation category. El Moto was recognized in March 2025. In a September release, the company describes construction during 2026 and mine-ramp development from July. It targets first production in the first quarter of 2029. The developer’s first-production target remains Q1 2029, following its 2026 inauguration and construction disclosures. [S04, S05, S24]

The proposed EU APT Tungsten Plant received recognition in the October list. On 9 October, the developer stated that it was evaluating the project and holding initial discussions with potential technical partners. It expressly described the final investment decision as pending. The possible processing of part of El Moto’s material is a future development proposition, conditional on evaluation and a subsequent decision. [S08, S09, S25]

The paired cases place the same designation across two distinct industrial stages. The same designation covers a mine reporting construction and a processing proposal awaiting investment approval. Treating both as operating capacity would erase a commercially important difference.

It also identifies a source-reading hazard. A project’s own communications may use verbs such as inaugurates or describe a future integrated chain. Current production and an operationally committed connection between facilities require separate evidence. The analysis therefore draws on the release’s stated operating milestones and forward-looking development conditions.

El Moto’s development history predates recognition, while the APT proposal remains at an early commercial stage. Causal attribution and future project outcomes require additional evidence. El Moto had lengthy pre-existing development work, and the APT decision was recent at the cutoff. The conclusion concerns the designation’s documented administrative effect and each project’s reported development status.

PUBLIC CAPITAL

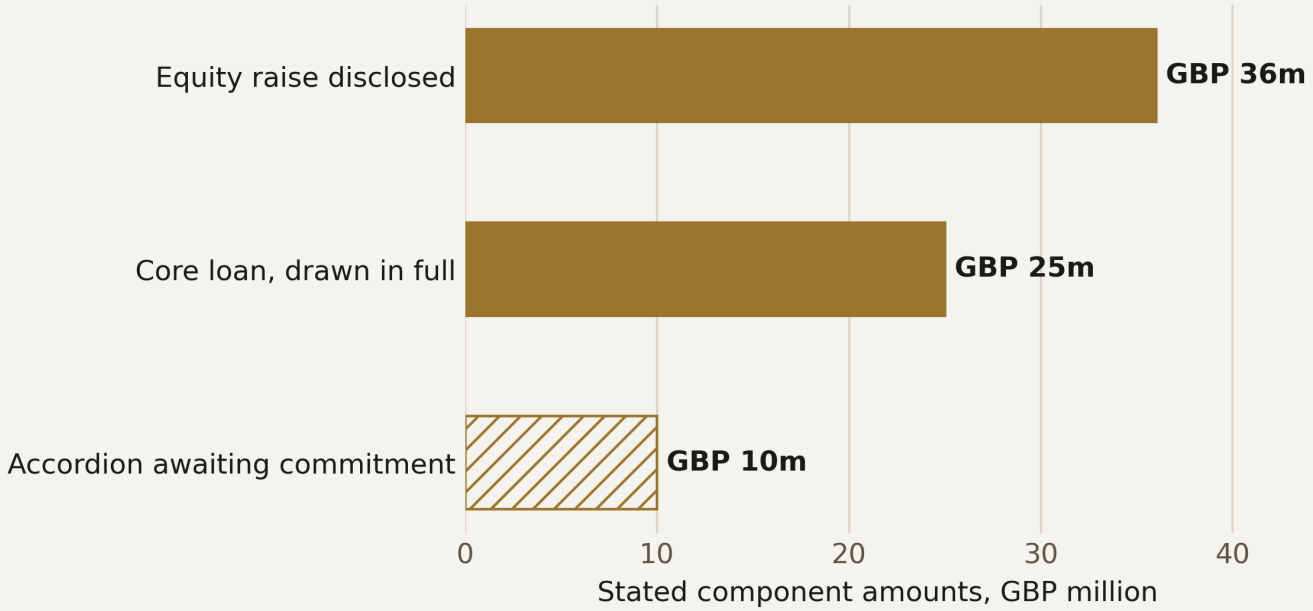
The UK £71 million package combines equity and contingent lending

The NWF announcement presents an investment of up to GBP 71 million for the Hemerdon restart. It describes GBP 36 million of equity and up to GBP 35 million of lending. The issuer's detailed terms split lending into a GBP 25 million core facility and a GBP 10 million accordion contingent on separate commitment. Later financial statements retain that distinction. [S14, S20, S31]

The maximum headline thus includes a contingent enlargement of the loan. The headline records potential financing capacity. Actual payment evidence is established separately. The government also obtained a separate opportunity to negotiate purchases of up to 50% of specified production, with valuation dependent on any subsequent agreement. The commercial terms and monetary value of that prospective procurement arrangement require an executed agreement.

SMPT's parent–component relationships make the most immediate error visible: adding the GBP 71 million package, GBP 36 million equity and GBP 35 million lending would produce GBP 142 million from one headline. The reproduction scripts identify the parent and its parts; the exhibit then uses supplemental issuer evidence to split the lending. This reconciliation presents the components of a single financing package and their distinct economic characteristics.

Exhibit 4. The up-to-GBP 71m package includes contingent GBP 10m lending



Sources: S14, S20 debt terms; S31 PDF pp. 39 and 67. GBP 36m equity, GBP 25m core loan and GBP 10m accordion reconcile the headline; they are different instruments and statuses. The accounts explicitly confirm full drawdown of the GBP 25m core loan. Procurement negotiation is separate and unpriced.

FINANCIAL EXECUTION

Tungsten West confirms the £25 million core-loan drawdown

The financial evidence advances beyond the NWF announcement. Tungsten West's FY2026 accounts, approved on 2 September 2026, disclose that the GBP 25 million NWF loan had been drawn in full by the date of the accounts. The statement appears in the accounting-policy going-concern discussion and again in note 36 on post-balance-sheet events. Those are more specific observations than a financing headline or an expected facility. [S31, PDF pp. 39 and 67; printed pp. 37 and 65]

The timing requires care. The reporting year ended on 31 March 2026. The NWF funding occurred subsequently. It would be wrong to read the year-end cash-flow statement as if it already included the August NWF transaction, or to date the loan drawdown to 31 March. The accounts establish full drawdown by their 2 September approval date. Establishing the transfer day requires transaction-level evidence.

The accounts also describe an approximately GBP 36 million equity raise. The October investor page lists NWF as a 7.4% shareholder. The issuer disclosures support the documented equity relationship. Bank-level confirmation remains a separate verification step. The contingent accordion remains separate. The core loan's disclosed drawdown establishes liquidity and a repayment obligation; expenditure allocation requires subsequent cash-use evidence. [S18, S20, S31]

The August issuer announcement proposed a secured 366-day facility at SONIA plus 5.5 percentage points, rising by one percentage point per quarter. The accounts also discuss the short repayment horizon. The proposed terms are treated as announced terms, subject to verification against definitive executed agreements. Its GBP 88 million sources-and-uses plan includes GBP 18 million of bridge-loan repayment and GBP 15 million of forecast revenue. Public capital combines equity and debt exposure. Project expenditure and grant-equivalent value require separate accounting. Cash generation and refinancing remain material execution risks. [S20, debt terms and appendix; S31, PDF pp. 40, 73]

A financing package can ease a liquidity constraint. Incremental investment attributable to that intervention remains a counterfactual question. Evaluating public value would require the financing alternatives, expected losses and returns, use of funds, and the security benefit purchased relative to other uses of public capital. Estimating additional investment and grant-equivalent cost would require a separate financing model.

OPERATIONAL EVIDENCE

Hemerdon reports processing amid an earlier restart timeline

On 15 September, Tungsten West reported that Hemerdon was processing ore and progressing through operational ramp-up. The release includes an NWF chief executive’s account of seeing ore being processed at the site. The September issuer disclosure provides evidence of processing activity following the August funding announcement. Independent throughput measurements would provide a further level of verification. [S21]

The documented milestone is commissioning-stage processing. The accounts describe phased commissioning and operation: an initial fines-gravity stage drawing on existing stockpiles and soft rock, further coarse-circuit work, and later completion of the permanent plant. The target of full-scale production in Q1 2027 remains a target at this cutoff. Sustained throughput, saleable concentrate output and deliveries to end users are subsequent operating and commercial milestones. [S31, PDF p. 2; S14]

The 25 August issuer announcement says concentrate had already been produced during the preceding month. The accounts discuss phased commissioning, with July and August references for different stages. Some industrial activity therefore preceded the announcement. Anticipated financing, prior negotiations and support for subsequent ramp-up remain plausible causal channels. Attribution requires estimating the alternative production path under different financing conditions. [S20, S31]

Existing infrastructure, earlier private fundraising, management decisions, tungsten prices and project economics may also contribute. Separating their marginal effects requires a counterfactual operational and financing model. Causal tonnage and fiscal cost per incremental tonne remain future analytical extensions.

Hemerdon supplies documentary evidence of public-loan drawdown and company-reported processing. The subsequent commercial agreement strengthens the route-to-market evidence, considered next. Sustained throughput, recovery, quality and completed deliveries are still needed to establish reliable supply.

CONTRACTS AND REMAINING EXPOSURE

Elmet signs an eight-year Hemerdon tungsten offtake

On 23 September, Tungsten West announced a binding eight-year Elmet offtake for 1,000 tonnes a year of contained WO3 at variable European APT-linked prices. This is contract evidence beyond project recognition and commissioning. Delivery records and realized revenue are subsequent milestones. [S35, RNS 8695V, further information]

Elmet’s 22 September company announcement, furnished as SEC Exhibit 99.1, corroborates the relationship and describes conversion through an allied refining network. It states an expectation of more than 1,000 tonnes; the dossier uses Tungsten West’s 1,000-tonne contractual description. The company release, furnished as SEC Exhibit 99.1, communicates issuer assertions. Industrial performance requires independent operating evidence. [S36]

The proposed government purchase arrangement remains distinct. The Elmet release provides for commercially reasonable efforts to process up to a further 500 tonnes annually for UK requirements, subject to commercial and delivery terms. The conditional processing provision carries a different level of contractual commitment from the binding offtake. The indicative contract value depends on realized volumes, pricing and contractual performance. [S35]

This changes the commercial question from whether any route to market has been announced to whether the contracted chain performs. Relevant evidence includes shipments, conversion availability, specifications, realized pricing and customer acceptance. APT-linked pricing leaves commodity-price exposure, while the short public loan raises a separate liquidity question.

The case illustrates an allied conversion route linking UK upstream material to a downstream commercial partner. UK extraction can connect to conversion elsewhere. This may improve supply options, but residual geographic, counterparty, feedstock and operating dependencies require their own assessment. Resilience gains require evidence of deliveries, conversion performance and supply continuity.

SUPPORT DECISIONS

JOGMEC records two tungsten conversion grant decisions

The Japanese comparison follows two linked but distinct administrative processes. SMPT records METI certification for Allied Material on 18 March 2026 and Japan New Metals on 7 September 2026. JOGMEC's programme page explains a subsequent grant application and decision process. Certification and payment occur at different stages of the support process. [SMPT event records; S12]

The current JOGMEC page lists Allied Material in its grant-decision table for FY2025, with approximately JPY 7.5 billion, and Japan New Metals for FY2026, with approximately JPY 2 billion. The current JOGMEC entry advances the documented status beyond the frozen SMPT row, which records the earlier announcement stage. The exact award date and disbursed amount require further programme documentation. [S12]

The METI certification page returned an access restriction during the review. The chronology attributes exact certification dates to the pinned SMPT records. Independent confirmation requires access to the current ministry listing. JOGMEC and recipient releases independently support the award and project descriptions used here. [S12, S13, S17]

The table keeps approximate grants apart from total investment plans. The programme's ceiling covers up to half the eligible undertaking cost. Actual subsidy intensity requires a verified eligible-cost denominator, which may differ from total announced investment. The analysis presents grant decisions and investment plans separately, preserving their different financial bases.

Exhibit 5. Japanese awards support future conversion expansions

Recipient	Grant award	Total plan	Capacity target	Operating target
Allied Material	~JPY 7.5bn	~JPY 15.9bn	~1.5x powder / carbide capacity	First half FY2028
Japan New Metals	~JPY 2bn	~JPY 8bn	2,400 tonnes/year WO3-equivalent APT capacity	April 2029

Sources: S12 grant-decision rows 7 and 10; S13 PDF p. 1; S17. Amounts are approximate. Grant awards and investment plans have distinct financial bases. Capacity refers to future WO3-equivalent APT output; disbursement and industrial output require further evidence.

CONVERSION AND RECYCLING

Allied Material and Japan New Metals plan conversion expansions

Allied Material’s 9 April release announces approximately JPY 15.9 billion for a new plant and related equipment, aiming for roughly 1.5 times existing tungsten and tungsten carbide powder capacity. It describes a site secured near Toyama Works and an operating target in the first half of FY2028. The reported milestone is equipment and site preparation; construction start and new-facility operation remain future verification points. The site and investment plan establish project preparation, with the expansion’s operating stage scheduled for future years. [S13]

Japan New Metals’ parent, Mitsubishi Materials, identifies a different conversion problem. Its September announcement describes approximately JPY 8 billion for equipment at Akita, including hydrometallurgical facilities that produce APT from scrap. The target is approximately 2,400 tonnes per year of APT capacity expressed as WO3 equivalent, about twice the current capacity. Operations are scheduled for April 2029, with installation planned sequentially from the fiscal year ending March 2027. [S17]

That target must retain its unit and denominator. The 2,400-tonne figure is total planned annual APT capacity on a WO3-equivalent basis. Incremental tungsten-metal content and achieved production require separate calculation and evidence. The company also describes an ambition to replace approximately 1,200 tonnes per year of imported intermediates on the same WO3-equivalent basis. The reported capacity remains a company target; observed import substitution requires subsequent production and sourcing evidence.

The two projects target tungsten conversion and recycled feedstock processing, distinct from upstream mining. Powder production still requires appropriate inputs. Scrap-to-APT expansion also requires recoverable scrap and process performance. The commercial value of these initiatives therefore depends on feedstock, yield, equipment commissioning and customer requirements, as well as public support.

The new expansions have future operating targets; their performance can be evaluated as commissioning and production records emerge. Their disclosed operating dates lie in 2028–29. The more immediate questions are whether awards translate into cash, installation and commissioning on schedule. A later outcome study should distinguish existing plant production from the incremental output of the supported expansions.

BREADTH AND COVERAGE

The EU adds APT, mining and recycling projects to its strategic list

The October EU annex adds three tungsten-related projects with distinct functions. Borralha is identified as extraction in Portugal, the EU APT Tungsten Plant as processing in Spain, and REW-ITA green HUB as recycling in Italy. The last also names copper and cobalt. Designation counts include a multi-material undertaking; tungsten-specific capacity requires project-level technical estimates. [S08]

Borralha’s promoter describes recognition as development and financing support, with funding decisions and environmental permitting remaining separate requirements. The release presents mineral-resource information and ongoing development. Resource estimates describe geological potential. Current production from the recognized project requires contemporary operating records. [S29]

REW-ITA’s promoter describes a gradual increase in capacity between 2026 and 2030 and work to consolidate recycling processes and explore additional approaches. The release describes European Commission support for some pilot initiatives. Project-specific award amounts and incremental operating output require further disclosures. The analysis records the support description and leaves the financial-allocation field for a verified project-specific award. [S26]

The same discipline applies to P6 Metals in the March list. The designation source supports project identity and industrial function. Current operating status requires a targeted project-level evidence review. It would be misleading to classify the project as inactive merely because the reviewed evidence is thinner. P6 remains an explicit evidence gap in the comparative matrix. [S04, S05]

These cases show both the value and the limit of a designation register. It can locate industrial-policy attention across stages and preserve the documentary basis for recognition. Determining industrial readiness requires separate investigation. Its yield varies with project maturity, source access and disclosure quality. A blank execution field directs the next targeted verification.

EVIDENCE AS OF 3 NOVEMBER 2026

Implementation status differs across seven selected projects

The matrix separates capital and investment decisions, physical work, operation and commercial evidence. The table separates administrative recognition, financing, physical development and operating evidence into independent dimensions. “Pending verification” identifies a milestone for which the reviewed sources provide insufficient confirmation. Its real-world status requires further evidence. New expansion output remains distinct from an existing company’s operations.

Exhibit 6. Eight projects have different kinds of documented progress

Project	Capital / investment	Physical progress	Operations / market
Hemerdon	GBP 36m equity raise; GBP 25m loan drawn	Phased restart / commissioning reported	Processing reported; Elmet contract; deliveries and steady state unverified
El Moto	No public payment established in reviewed set	Construction and ramp development reported	Q1 2029 first-production target
EU APT Tungsten Plant	No final investment decision reported	Evaluation / technical-partner discussions	No operation of proposed plant established
P6 Metals	Pending verification in reviewed set	Pending verification in reviewed set	Current operation not established in reviewed set
Borralha	Designation is not a funding award	Development / resource work described	New industrial output not established
REW-ITA green HUB	Pilot support mentioned; no attributable amount	2026-2030 expansion programme reported	Incremental expansion output not quantified
Allied Material	Approx. JPY 7.5bn grant decision; payment unknown	Site secured; plant and equipment plan	First-half FY2028 operating target
Japan New Metals	Approx. JPY 2bn grant decision now listed	Akita equipment investment approved; installation planned	April 2029 expansion operating target

Sources: implementation.csv and locators in the workbook. “Pending verification” identifies milestones requiring additional documentation. New expansion output is distinguished from existing operations. Labels show different evidence categories. As of 3 November 2026.

SKEPTICAL ASSESSMENT

Project maturity and disclosure quality shape the comparison

Coverage is unequal. The UK case has financial statements and detailed securities disclosures. The Japanese cases have grant lists and equipment announcements. Several EU entries are documented primarily through designation decisions and promoter statements. Greater disclosure can increase observed evidence density independently of underlying project progress. The comparison reports evidence categories for each instrument and project.

Project maturity is unequal. Hemerdon is a restart using existing infrastructure, El Moto is in construction with a 2029 target, and the APT plant is awaiting an investment decision. A single-date output comparison would favor projects that entered the policy period at a more advanced development stage.

Objectives are unequal. China’s licensing obligation manages controlled transactions; designation supports project development; finance changes access to capital. A common test of tonnes produced would omit important effects of licensing and administrative support. A common test of announcements would ignore whether assets or material actually emerge.

The UK documents establish drawdown, reported processing and a commercial offtake. Pre-announcement activity prevents simple attribution of the whole restart to the August announcement; anticipation could still matter. The documents establish execution milestones; the incremental effect of intervention requires a counterfactual. [S20, S21, S31, S35]

The evidence supports a differentiated answer to the central question. Some interventions have reached documented financial execution and industrial activity. Others remain at recognition, award decision or planned expansion in the reviewed sources. Estimating aggregate supply reliability, relative government effectiveness and causal necessity of public support requires harmonized output data and counterfactual analysis.

The conclusion addresses the evidentiary status of these selected interventions at the research cutoff. It identifies where the empirical burden lies. Evaluation must connect the instrument’s actual mechanism to a measurable industrial or transactional consequence, preserve the counterfactual question, and distinguish missing disclosure from missing activity.

POLICY AND COMMERCIAL USE

Financiers and buyers need evidence of output, delivery and conversion

For public financiers, disclosure can make the difference between a headline and an evaluable intervention. Separate core facilities from contingent enlargements; publish award, agreement and drawdown dates; and identify whether reported output comes from the supported asset or an existing line. An annual accounts note can materially change the understanding of execution, as the NWF loan disclosure demonstrates. [S31]

For industrial-policy teams, designation programmes would be easier to evaluate with project-level reporting on permitting progress, financing closure, construction and incremental capability. Recognition has policy value even before production. Its contribution should be measured through the mechanism it targets: permit processing, investment mobilization or industrial execution as appropriate.

For defense and manufacturing buyers, qualified supply requires product specifications, verified conversion, delivery schedules and customer acceptance. The selected projects provide different products at different dates. Procurement diligence should identify feedstock, conversion access, quality specifications, commissioning progress and the arrangements under which material reaches users. A domestic mine can still depend on conversion elsewhere; a domestic conversion expansion can still depend on imported inputs.

For government-affairs and strategy advisers, the relevant question is which constraint the instrument addresses. A loan may reduce a funding gap, a designation may improve administrative coordination, and export licensing may alter transaction uncertainty. Advice based on a single count of policies or “secured” capacity risks recommending action against the wrong constraint.

For investors and research users, the next evidence request should be specific. Hemerdon calls for sustained production and delivery data. Allied Material and Japan New Metals call for payment, equipment and commissioning updates against their announced schedules. The proposed APT plant calls for an investment decision and financing evidence. The licensing case calls for item-aligned transaction and implementation data.

These implications guide due diligence, sourcing and research priorities; forecasts of prices, returns and compliance require separate models and legal analysis. The commercial contribution of the report is to make the missing evidence actionable. The framework prioritizes observable project and transaction evidence over synthetic scores.

THREE WORKED RESEARCH DEMONSTRATIONS

SMPT connects policy instruments, financing and industrial projects

The frozen export contains 66 events, 189 sources, 102 projects, 59 designations, 122 organizations, 96 financial rows and 45 control rows. Exact tungsten tags yield 19 events, nine projects, eight finance rows, two controls and eight designations. Scoping issuers to the four systems gives 11 events, eight linked projects, six financial rows, two controls and eight designations. These counts describe linked record types in the frozen SMPT selection and its existing coverage.

The UK trail follows the event to a financing parent, its parts and the Hemerdon project, preventing package double-counting. Issuer terms and accounts refine that trail into a core loan, contingent accordion and documented drawdown. Separate commercial disclosures add the Elmet agreement. Claim-level SMPT IDs and source locators are supplied in the workbook. [S14, S20, S31, S35, S36]

The Japanese trail connects certification, a grant row and a project to JOGMEC's decision table and the parent-company expansion release. The current table advances Japan New Metals beyond the frozen announcement-only status. The relationships guide supplemental investigation, whose findings remain separately documented from the frozen public dataset. [S12, S17]

The platform's practical contribution is the preservation of relationships, classifications and source locators. The researcher must still verify changes, interpret documentary language and challenge causal claims. The structured model supplies the investigation map; source review produces the additional project-level findings.

Exhibit 7. Three source-to-conclusion trails used in this study

Starting point	Research path	Result
China licensing / EU recognition	Event → control or designation → source item / annex	Legal obligations call for distinct implementation measures
Hemerdon financing	Event → package / parts → project → issuer terms / accounts	Avoid package duplication; identify the core loan actually drawn
Japan New Metals	Certification → grant / project → current JOGMEC / MMC release	Award status has advanced; Akita expansion remains future capacity

Sources: S01, S08, S09, S12, S14, S17, S20, S31. Full SMPT IDs and source locators are in claims.csv, the raw selection and the evidence workbook. The original public dataset remains frozen.

METHOD AND RESEARCH RECORD

Dataset, sources and claim-level reproduction

Snapshot. The research archive preserves the genuine public dataset.json download, the exact pinned seed collections, export code, methodology page source and styling source. The manifest records source paths, retrieval timestamps, byte counts and SHA-256 hashes. The live export identifies a 2026-11-03 dataset-generation date. Retrieval timing is recorded separately in the evidence register.

Selection. select.py first selects exact tungsten tags across events, financial rows, controls, designations and projects. build_analysis.py narrows issuing jurisdictions to the four selected systems and follows project IDs through scoped finance and designation rows. The broad material subset is retained for audit. The comparison uses 11 events, eight projects, eight designations, six raw financing rows and two control clauses. Hemerdon retains one physical-project identity across UK financing and EU designation records.

Transformation. Counts are raw row counts, deliberately distinct from SMPT's own package-folded capital summaries. Financial amounts retain original currencies and qualifiers. The UK core loan and accordion are extracted from the issuer's detailed terms and reconciled to the package. Japanese grant and investment-plan amounts are shown separately. All currencies and technical capacities retain their stated units. The analysis reports instrument- and project-level observations.

Evidence. The workbook connects each principal claim to its source statement, pinned record, analytical interpretation and next verification requirement. Project status labels are editorial summaries based on the cited records; platform taxonomy remains unchanged. Exact grant-decision days and some notification dates remain unknown. Existing and proposed capacity appear as distinct industrial stages.

Figures. Seven main exhibits use reviewed inputs and explicit figure definitions. The original raw-count graphic is retained only as an archive asset; its counts appear in the method and workbook. SVG, PNG, PDF, CSV inputs and scripts are included. Offline figure regeneration uses reviewed_analysis.json. Original-document extraction requires re-retrieval of the source files and may be limited by access or later revisions. Third-party evidence appears through bibliographic citations and external source links.

Limits. The purposive comparison covers selected instruments and uneven levels of disclosure across the four jurisdictions. The verification method consists of legal instruments, corporate disclosures, financial statements and official programme records. Causal additionality and trade effects require separate project- and transaction-level models. Transaction-specific use requires original-language legal review. This is an authored independent research dossier, with external expert review remaining a potential next quality-assurance stage.

SOURCE REGISTER

References

SMPT dataset and methodology. Public export: https://strategic-materials-policy-tracker.vercel.app/api/export/dataset.json. Repository commit 4fc59c7ea38ca788eb7537353274a953c90e13eb. The evidence workbook supplies exact record IDs, URLs, access results and retrieval metadata.

S01. MOFCOM/GACC Announcement No. 10 of 2025 (tungsten et al.). Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

S02. MOFCOM announcement of a special campaign against strategic-mineral smuggling, 9 May 2025. Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

S04. Annex to Commission Decision C(2025) 1904 final: list of Strategic Projects. Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

S05. Commission Decision of 25.3.2025 recognising certain critical raw material projects as Strategic Projects under Regulation (EU) 2024/1252 (C(2025) 1904 final). Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

S06. Annex to Commission Decision C(2025) 3491 final: list of Strategic Projects in third countries and overseas countries or territories. Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

S07. Commission Decision of 4.6.2025 recognising certain critical raw material projects located in third countries and in overseas countries or territories as Strategic Pro.... Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

S08. Annex to Commission Decision C(2026) 8051 final: list of Strategic Projects. Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

S09. Commission Decision of 9.10.2026 recognising certain critical raw material projects as Strategic Projects under Regulation (EU) 2024/1252 (C(2026) 8051 final). Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

SOURCE REGISTER

References (continued)

S11. METI critical-minerals certified supply-assurance plans. The current ministry page returned HTTP 403. Exact certification dates rely on the frozen SMPT records pending a fresh ministry-page verification.

S12. JOGMEC critical-minerals grant programme and grant-decision list. Current grant-decision rows 7 and 10, FY2025 and FY2026. Exact calendar decision days and disbursements require further programme records.

S13. Allied Material: tungsten capacity expansion and new plant, 9 April 2026 (Japanese original). Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

S14. National Wealth Fund backs Tungsten West to unlock critical UK tungsten supply. Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

S17. Mitsubishi Materials Japan New Metals expansion, 9 September 2026. Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

S18. Tungsten West investor information. Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

S19. MOFCOM export-control list copy of Announcement No. 10. Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

S20. Tungsten West NWF terms, issuer RNS mirror. Issuer RNS 9611R, 25 August 2026, read in a public mirror. Canonical LSE page returned metadata and an investor declaration wall. Debt terms and project update.

SOURCE REGISTER

References (continued)

S21. Tungsten West processing starts, issuer RNS mirror. Issuer RNS 7152U, 15 September 2026, read in a public mirror. Operating statements are company-reported; independent throughput measurement would provide a further verification tier.

S24. Abenojar El Moto construction, 18 September 2026. Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

S25. Abenojar APT Plant designation, 9 October 2026. Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

S26. OMCD REW-ITA project disclosure, 15 October 2026. Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

S27. EU official permitting guide. Primary issuer or official source, retrieved 3 November 2026. See evidence workbook for claim-level locators.

S29. Borralha promoter designation and implementation disclosure. Promoter disclosure used for development description. The official annex confirms project recognition; resources concern geological potential.

S30. Official Spanish reproduction of EU Decision 2026/2349. Official Spanish reproduction confirms OJ publication of Decision 2026/2349 on 22 October 2026. Exact notification date remains unresolved.

S31. Tungsten West audited annual report FY2026, including subsequent events. Accounts approved 2 September 2026. PDF pp. 39 and 67 (printed pp. 37 and 65) disclose full loan drawdown; PDF pp. 2 and 39 describe phased commissioning. The financing occurred after the 31 March fiscal year end.

SOURCE REGISTER

References (continued)

S32. USGS Mineral Commodity Summaries 2026 tungsten pages 200 and 201. USGS estimates for 2025, tungsten chapter printed pp. 200–201. The figures report mine tungsten content; conversion capacity and buyer exposure use different measures.

S33. Regulation EU 2024/1252 Article 11 official original text. Article 11, especially paragraphs 1–4 and 6; official legal text independently consulted during review.

S34. TFEU Article 297 notification and effect. Article 297(2), official indexed excerpt. Direct current HTML response unavailable. Exact October notification date unresolved.

S35. Tungsten West RNS 8695V Elmet offtake 23 September 2026. Issuer RNS 8695V, 23 September 2026, further information and UK strategic supply sections. Deliveries require later shipping and customer receipt records.

S36. Elmet company release furnished as Exhibit 99.1 under Item 7.01, 22 September 2026. Counterparty company release dated 22 September 2026, furnished as SEC Exhibit 99.1 pursuant to Item 7.01. Its issuer statements remain subject to independent industrial-performance verification.